

The effect of electric charge density of polyacrylamide (PAM) on properties of PAM/cellulose nanofibril composite films

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Abstract Poly(acrylamides) (PAMs) with various anionic and cationic charge densities from -1.64 to $+0.98$ meq/g but similar molecular weights were synthesized through radical polymerization. One of the aqueous PAM solutions and an aqueous dispersion of 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO)-oxidized cellulose nanofibril (TOCN) were mixed at various PAM/TOCN weight ratios, and self-standing PAM/TOCN composite films were prepared by casting and drying. Because nonionic and cationic PAM molecules may have had some attractive interactions with anionic TOCN elements, the visual appearances and particle sizes of the mixtures varied depending on the charge densities of PAMs. In particular, PAM molecules with electric charge densities with $+0.54$ and $+0.98$ meq/g formed polyelectrolyte complexes with anionic TOCN elements, resulting in formation of large agglomerates in the aqueous mixtures. The PAM/TOCN composite films at weight ratios of 10/90 and 25/75 had higher Young's moduli and tensile strengths than those of 100 % TOCN film, when

PAMs with charge densities from -1.64 to $+0.54$ were used. The PAM/TOCN composite films at the weight ratio of 10/90 had elongations at break similar to that of 100 % TOCN film, resulting in higher work of fracture values or ductile properties, whereas the PAM/TOCN composite films at the weight ratio of 25/75 were rather brittle.

Keywords TEMPO · Cellulose nanofibril · Poly(acrylamide) · Electric charge · Composite film · Mechanical properties

Introduction

Native plant cellulosic materials have extremely tough structures consisting of highly crystalline cellulose microfibrils with ~ 3 nm thickness as the smallest element (next to cellulose molecules themselves). Recently, efficient processes to downsize plant cellulose fibers with or without pretreatment to nano-size elements using new mechanical disintegration methods have been developed, which have opened new fields of bio-based nanomaterial science and technologies associated with the global green chemistry boom for establishment of a sustainable society. Moreover, wood celluloses produced as bleached chemical pulps in the pulp and paper industry, using conventional pulping and bleaching technologies can be used as a resource for nano-sized

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celluloses, which is encouraging for the present day paper industry to expand to a new high-tech biorefinery industry.

Nano-sized celluloses are classified as nanocrystalline celluloses (NCCs) and nanofibrillated celluloses (NFCs) with low and high aspect ratios, respectively (Habibi et al. 2010; Klemm et al. 2011; Isogai 2013). NCCs are prepared from wood pulps by acid hydrolysis with, for example, 64 % sulfuric acid (Dong et al. 1998; Habibi et al. 2010). NFCs are obtained from wood pulps with or without chemical pretreatment through repeated mechanical grinding or high-pressure homogenization in water (Klemm et al. 2011; Isogai et al. 2011). In most cases, NFCs have network structures without complete individualization of the wood cellulose microfibrils. This may be advantageous for removing the water (abundantly present in aqueous NFC dispersions) when making dried films or sheets using a filtration process under vacuum similar to papermaking. Some of the self-standing NFC films had high mechanical strengths, optical transparency, thermal stability, and gas-barrier properties (Henriksson et al. 2008; Siró and Plackett 2010; Sehaqui et al. 2010; Liu et al. 2011).

We have reported that 2,2,6,6-tetramethylpiperidine-1-oxy radical (TEMPO)-mediated oxidation used as a pretreatment can efficiently convert wood cellulose fibers to completely individualized TEMPO-oxidized cellulose nanofibrils (TOCNs) with homogeneous widths of ~ 3 nm dispersed in water (Saito et al. 2006, 2007; Isogai et al. 2011; Isogai 2013). Anionic sodium carboxylate groups are densely and regularly present on TOCN element surfaces, with the result that osmotic effects and electrostatic repulsion efficiently work between the anionically charged TOCN elements both during mechanical disintegration treatment in water and in the subsequent dispersion state (Okita et al. 2010; Hirota et al. 2010; Isogai et al. 2011). The TOCN elements with or without surface modification maintained sufficiently nano-dispersed states in matrix polymers such as poly(styrene), poly(lactic acid), hydroxypropyl cellulose, and poly(vinylalcohol), resulting in efficient improvement of mechanical and thermal properties of the transparent and flexible composite films with addition of small amounts of TOCN (Johnson et al. 2009; Li et al. 2010; Fujisawa et al. 2012, 2013; Endo et al. 2013).

Poly(acrylamides) (PAMs) are water-soluble synthetic polymers, and have been used primarily as dry

strength additives of, for instance, in making paper-board for corrugated board production. Because molecular weight, degree of branching, and charge density of PAMs can be controlled by controlling the synthesizing conditions, various and tailor-made PAMs effective to particular papermaking conditions can be designed. The abundant amide groups of PAMs are expected to reinforce inter-TOCN bonds through hydrogen bonds between TOCN elements and PAM molecules in preparation of transparent and flexible PAM/TOCN composite films.

In a previous paper, aqueous solutions of anionic PAMs with various molecular weights and degrees of branching were mixed with an aqueous TOCN dispersion at various weight ratios of PAM/TOCN, and self-standing PAM/TOCN composite films were prepared by casting and drying of homogeneous mixtures (Kurihara and Isogai 2014). The composite films had clearly higher Young's modulus and tensile strength than 100 % TOCN film, when the PAM/TOCN weight ratio was 25/75, showing the mechanical reinforcement ability of PAM molecules in TOCN matrix. Furthermore, the mechanical properties of the PAM/TOCN composite films were improved by controlling molecular mass and branching degree of the PAM. In this study, PAMs with different electric charge densities were synthesized, and used to prepare PAM/TOCN composite films. Optical, tensile and other fundamental properties of the composite films were studied in terms of electric charge density of PAMs.

Experimental

Materials

A commercial softwood bleached kraft pulp (SBKP) for papermaking (Nippon Paper Co., Ltd., Tokyo, Japan) was used as the wood cellulose after demineralization by soaking in 0.1 M HCl at room temperature for 2 h (Shinoda et al. 2012). A commercial 50 % aqueous solution of acrylamide product (Mitsui Chemicals, Tokyo, Japan) was used. A 98 % aqueous solution of acrylic acid (Nippon Shokubai Co., Ltd., Osaka, Japan) and *N,N*-dimethylaminopropyl acrylamide (DMAPAA) (KJ Chemicals Co., Tokyo, Japan) were used as anionic and cationic monomers, respectively. Other reagents and solvents were commercial

products (Wako Pure Chemical, Osaka, Japan) and used as received.

Preparations of aqueous TOCN dispersion and PAM solution

A TEMPO-oxidized cellulose was prepared from SBKP using the TEMPO/NaBr/NaClO process in water at pH 10 and room temperature and the successive oxidation with NaClO₂ in water at pH 4.8 according to a previously reported method (Kurihara and Isogai 2014). The TEMPO-oxidized cellulose thus obtained had a carboxylate content of 1.30 mmol/g, as determined by conductivity titration (Kurihara and Isogai 2014). Aqueous TOCN dispersions (0.15 % w/w) were prepared from aqueous TEMPO-oxidized cellulose slurries by mechanical disintegration treatment using a blender-type homogenizer (Excel Auto ED-4, Nissei, Japan) at 15,000 rpm for 5 min and an ultrasonic homogenizer at 19.5 kHz and 300 W output power with a 26 mm probe tip (US-300T, Nissei, Japan) for 6 min. Any unfibrillated or partly fibrillated fraction, present in only a small amount, in the dispersion (<5 %) was removed through centrifugation at 12,000g for 10 min.

PAMs were synthesized in water through free-radical polymerization from acrylamide and one of the ionic vinyl monomers with a small amount of a branching agent. In all PAM synthesis, Milli-Q water was used. The detailed procedure to synthesis PAMs was described in a previous paper (Kurihara and Isogai 2014). All PAMs were synthesized to have almost the same molecular weight but different electric charge densities, by controlling polymerization conditions and weight ratios of monomers. The solid contents of the aqueous PAM solutions were adjusted to 10 % w/w, which was diluted with water to 0.15 % w/w just before addition to the 0.15 % TOCN dispersion.

Preparation of PAM/TOCN composite films

To the 0.15 % w/w TOCN dispersion, a designed amount of the 0.15 % w/w PAM solution was slowly added under stirring to adjust the weight ratio of PAM/TOCN to either 10/90 or 25/75. The self-standing PAM/TOCN composite films were prepared from the aqueous PAM/TOCN mixtures according to the procedure described in the previous paper (Kurihara and Isogai 2014). The PAM/TOCN composite films

thus obtained, approximately 10 µm thick, were conditioned at 23 °C and 50 % relative humidity (RH) for more than 1 day.

Analyses

The molecular weight parameters of PAMs were measured by size-exclusion chromatography (SEC) with right angle-laser light scattering (RALLS). The detailed SEC conditions were described elsewhere (Kurihara and Isogai 2014). The colloidal titration was adopted to determine electrical charge densities of PAMs, using a Mutek PCD-04 streaming current detector (Germany) with either 0.0025 N poly(diallyldimethylammonium chloride) or potassium poly(vinyl sulfate). The particle sizes of 0.15 % PAM/TOCN mixtures were measured at 25 °C by dynamic light scanning (DLS) (i.e. photon correlation spectroscopy) using an ELSZ-2 particle size analyzer (Otsuka Electronics Co., Ltd., Japan). Water was used in the dispersion medium, and a refractive index and viscosity of water were set to 1.3328 and 0.8878 mPa s, respectively. The thickness, moisture content at 23 °C and 50 % RH, light transmittance spectrum, and tensile properties of the composite films with approximately 10 µm thickness were obtained through the methods described in the previous paper (Kurihara and Isogai 2014); at least 10 film specimens were measured for each sample in the tensile test.

Results and discussion

Aqueous PAM/TOCN dispersions

Fundamental properties of anionic, nonionic, and cationic PAMs synthesized and used in this study are listed in Table 1; PAMs with different charge densities but almost the same weight-average molecular weights (M_w) and radii of gyration were synthesized by controlling monomer ratios and polymerization conditions. All PAM solutions were transparent, and PAM molecules in the aqueous 0.05 % PAM solutions at pH 7 had electric charge densities from -1.64 to $+0.98$ meq/g. The aqueous 0.05 % TOCN dispersion used in this study had, on the other hand, a charge density of -0.98 meq/g.

It has been reported that the self-standing PAM/TOCN composite films by casting and drying of

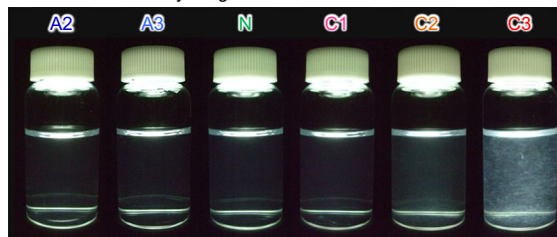
Table 1 Charge density and molecular weight of PAMs prepared in this study

PAM code	Charge	Charge density at pH 7 (meq/g)	M _w (M Da)	M _w /M _n	Radius of gyration (nm)
A1	Anion	−1.64	1.87	4.1	59.4
A2	Anion	−1.02	2.06	5.7	62.2
A3	Anion	−0.71	2.07	5.9	60.0
N	Nonion	−0.02	2.00	5.4	62.8
C1	Cation	+0.09	2.17	6.1	62.3
C2	Cation	+0.54	1.96	4.8	60.3
C3	Cation	+0.98	1.79	4.4	57.1

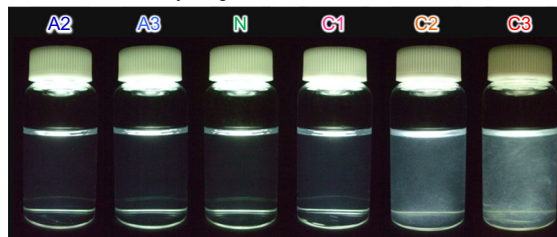
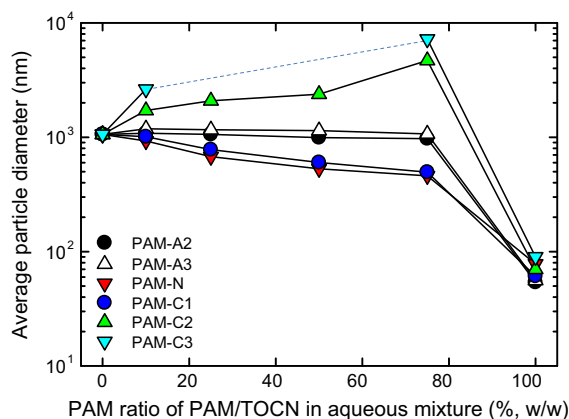
aqueous mixtures of an anionic PAM (A2 in Table 1) and TOCN at a weight ratio of 25/75 gave the highest tensile strengths (Kurihara and Isogai 2014). In this study, therefore, the weight ratios of PAM/TOCN were fixed to 10/90 and 25/75 to prepare PAM/TOCN composite films with PAMs having different electric charge densities. When the PAM/TOCN mixtures at weight ratios of 10/90 and 25/75 were prepared by adding the anionic and nonionic PAM solutions to the TOCN dispersion, the mixtures maintained transparent states. In the case of the addition of cationic PAMs, the mixtures became cloudy and some white agglomerates were observed in the mixtures. The more the cationic charge of the PAM, the more the mixture had agglomerates (Fig. 1).

The particle size of PAM/TOCN components in the mixtures was determined by DLS (Fig. 2). The mixtures were roughly categorized into three groups depending on electric charge density of PAMs; A2 and A3, N and C1, and C2 and C3. In the case of anionic PAM-A2 and PAM-A3 with large anionic charges, the anionic TOCN elements were sufficiently nano-dispersed in the mixtures similarly to the original TOCN dispersion regardless of the PAM mixing ratios, because electrostatic repulsions sufficiently work between these anionically charged components. When the nonionic PAM-N and slightly cationic PAM-C1 were used, the TOCN elements shrank a little in the mixtures or their particle sizes decreased as the PAM addition level was increased. Some physical interactions may have been formed between anionic TOCN elements and these PAM molecules in the mixtures by excluding electrostatic repulsions, probably resulting

PAM/TOCN=10/90 by weight



PAM/TOCN=25/75 by weight

**Fig. 1** Optical photographs of aqueous PAM/TOCN mixtures, that were prepared with various PAMs and TOCN at PAM/TOCN weight ratios of 10/90 and 25/75**Fig. 2** Relationship between PAM weight ratio of PAM/TOCN in aqueous mixtures and average particle diameter

from partial blocking of the electrostatic repulsion between TOCN elements in the mixtures.

In contrast, when the cationically charged PAM molecules, C2 and C3, were mixed with anionic TOCN elements, large amounts of PAM/TOCN agglomerates were formed, probably as polyelectrolyte complexes through ionic bonds between the components, resulting in the remarkable increase in the particle size. When the PAM-C3 with the highest cationic charge was mixed with TOCN at the PAM/

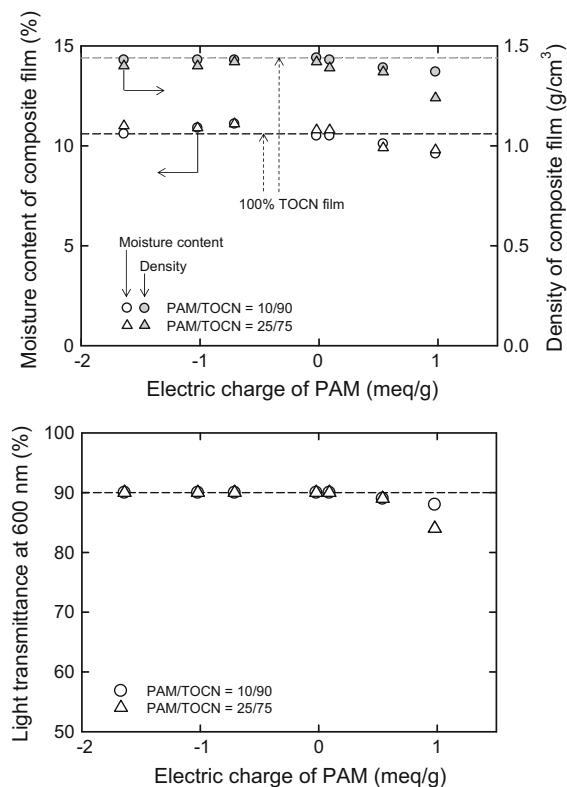


Fig. 3 Moisture content, density, and light transmittance at 600 nm of PAM/TOCN composite films, that were prepared with PAMs with different electric charge densities at PAM/TOCN weight ratios of 10/90 and 25/75

TOCN weight ratios of 25/75 and 50/50, their particle sizes could not be measured, because the mixtures turned to translucent gels by forming ionic bond networks between the PAM and TOCN components.

Properties of PAM/TOCN composite films

The self-standing PAM/TOCN composite films were successfully obtained from all the aqueous PAM/TOCN mixtures by casting and drying. Figure 3 shows the moisture content and film density at 23 °C and 50 % RH, and the light transmittance at 600 nm of the PAM/TOCN composite films. The PAM/TOCN composite films except those prepared with PAM-C2 or PAM-C3 had almost the same moisture contents, densities, and light transmittances as those of the 100 % TOCN film. When the highly cationically charged PAM-C2 or PAM-C3 was used, polyelectrolyte complexes were formed between cationic PAM molecules and anionic TOCN elements (Kurihara and

Isogai 2014). These polyelectrolyte complexes may have lower hydrophilic nature than the original ionic structures of PAMs and TOCN elements; the more ionic bonds are formed between anionic and cationic components, the more water molecules are released from the originally hydrophilic ionic sites so as to stabilize the agglomerates of polyelectrolyte complexes (Michaels 1965; Howard et al. 2000; Bernabé et al. 2005). Moreover, the formation of PAM/TOCN agglomerates large in size may have caused an increase in pore volume in the composite films, resulting in lower film density and lower light transmittance. Furthermore, the composite films prepared with PAM-C2 and PAM-C3 had relatively coarse surfaces, resulting from the formation of large agglomerates, whereas others had quite smooth surfaces similar to that of the 100 % TOCN film (Chen et al. 2003; Takahashi et al. 2009; Kurihara and Isogai 2014).

Figure 4 shows the tensile properties of the PAM/TOCN composite films prepared from aqueous PAM/TOCN mixtures at the PAM/TOCN weight ratios of 10/90 and 25/75 with PAMs having various electric charge densities. Both Young's modulus and tensile strength of the composite films, except those prepared with 25 % PAM-C3, were higher than those of the 100 % TOCN film. Thus, these properties of TOCN film were improved by including 10 and 25 % PAMs with a wide range of electric charge densities from -1.64 to $+0.54$ meq/g. In most cases, the composite films containing 25 % PAMs had higher values than those containing 10 % PAMs. Moreover, a careful observation of these results shows that the composite films prepared with PAMs with electric charge densities around zero gave higher Young's modulus and tensile strength than those prepared with PAM-A2. The strongly electrostatic repulsions presumably present between anionic TOCN elements and PAM-A2 molecules have not necessarily resulted in higher Young's modulus or tensile strength of the composite films. Instead, attractive interactions formed between the anionically charged TOCN elements and nonionic or weakly cationically charged PAM molecules brought about higher Young's modulus and tensile strengths of the PAM/TOCN composite films.

The elongations at break of the composite films consisting of 25 % PAMs of all electric charge densities were lower than that of 100 % TOCN film, resulting in lower work of fracture values; these composite films became rather brittle. In contrast, the

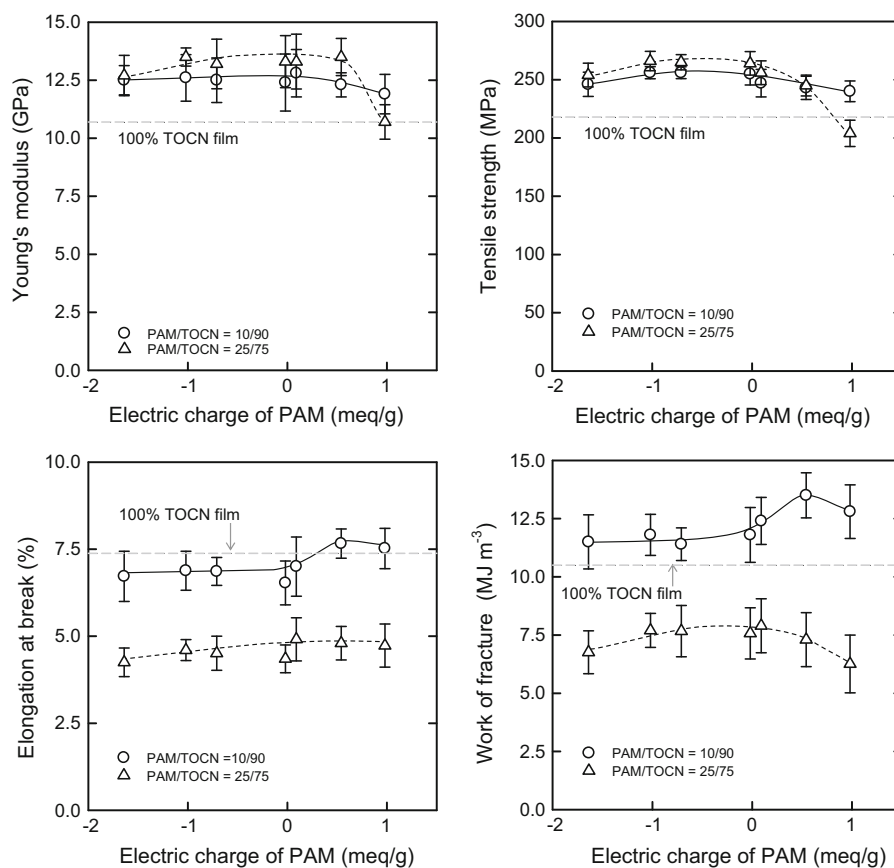


Fig. 4 Tensile properties of PAM/TOCN composite films, that were prepared with PAMs with different electric charge densities at PAM/TOCN weight ratios of 10/90 and 25/75. Error bars show SD

PAM/TOCN composite films prepared from the PAM/TOCN weight ratio of 10/90 with PAM-C2 or PAM-C3 had higher elongations at break, resulting in higher work of fracture values of $>12.5 \text{ MJ/m}^3$. Thus, the formation of PAM/TOCN agglomerates in the aqueous mixtures may have provided sufficiently ductile properties to the composite films; the electrostatically attractive interactions formed between cationic PAM molecules and anionic TOCN elements seem to have a positive effect on ductile properties. Consequently, the tensile properties of the PAM/TOCN composite films are controllable by controlling electric charge density of PAM used as the additive.

Tensile properties of PAM-A2/PAM-C3/TOCN composite films

Based on the results obtained in the previous section, we expected some synergistically improving effects

Table 2 Weight ratio of three components in PAM/TOCN composite films

	a	b	c	d	e	f
PAM-A2	0	25	24.5	24	20	0
PAM-C3	0	0	0.5	1	10	25
TOCN	100	75	75	75	75	75

on tensile properties of PAM/TOCN composite films using three components systems through additional electrostatically attractive interactions formed between anionic PAM molecules, cationic PAM molecules, and anionic TOCN elements. Table 2 shows the weight ratios of three components, PAM-A2, PAM-C3, and TOCN, of the composite films b–f, in which the weight ratios of PAMs/TOCN were fixed to 25/75. Moisture contents, film densities, and light transmittances at 600 nm were similar between the

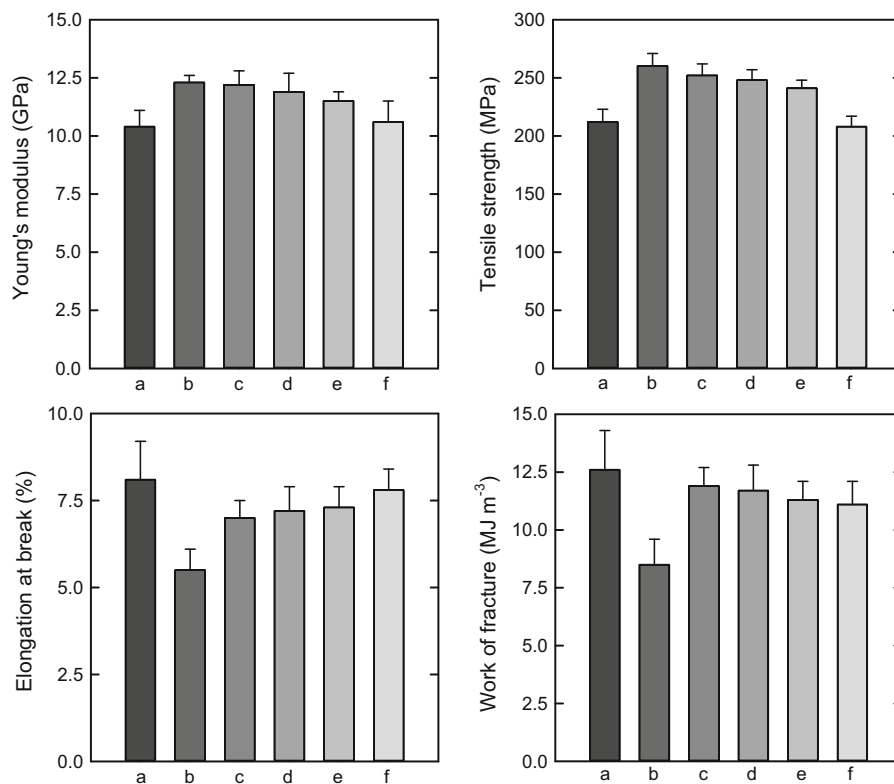


Fig. 5 Tensile properties of PAM/TOCN composite films, that were prepared with PAM-A2, PAM-C3, and TOCN with various weight ratios. Error bars show SD

composite films, and were 11.5–12.5, 1.39–1.45, and 87–90 %, respectively (Fig. 5).

The addition of PAM-C3 partially in place of PAM-A2 had almost no contribution to improvement of Young's modulus or tensile strength of the composite films. However, when 2 or 4 % PAM-A2 was replaced with cationic PAM-C3, the elongation at break and work of fracture value of the composite film was remarkably improved in comparison with that of PAM-A2/TOCN composite film. Thus, the replacement of PAM-A2 with even only a small amount of cationic PAM-C3 was effective to improve ductile properties of the composite films. Nevertheless, any synergistic effects were unfortunately not observed for these three component systems.

Conclusion

When the electric charge density of PAMs was controlled by controlling the weight ratio between nonionic acrylamide and either anionic or cationic

monomer in PAM synthesizing process, average particles sizes of TOCN elements or PAM/TOCN polyelectrolyte complexes in aqueous dispersions showed significant changes. When the PAM/TOCN weight ratios were 10/90 and 25/75 and the electric charge densities of PAMs ranged from -1.64 to $+0.09$ meq/g, the composite films had similar moisture contents, densities, and light transmittances at 600 nm. The Young's modulus and tensile strength of TOCN film were improved by co-production with PAMs with electric surface charges of -1.02 to $+0.54$ meq/g at the PAM/TOCN weight ratios of both 10/90 and 25/75. Although the PAM/TOCN composite films at the weight ratio of 25/75 became brittle, those at weight ratio of 10/90 had high elongations at break similar to that of 100 % TOCN film, resulting in high work of fracture values. Thus, the formation of polyelectrolyte complexes between cationic PAM molecules and anionic TOCN elements may provide ductile properties to the composite films. When 2 or 4 % of PAM-A2 was replaced with cationic PAM-C3, the PAMs/TOCN composite films had high Young's

modulus, tensile strength, and work of fracture. Because the molecular structures of PAMs are controllable, it is possible to design the PAM/TOCN composite films to have adequate mechanical properties for various end uses.

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